lec 10 16/9/21

RECAP

- FTP
- email
- DNS
  - ↳ How to design a DNS system?
    - Centralized system that holds one large file with mapping of hosts to IPs ???
      - Billions of records in the file
      - Several trillions of requests/day
      - Requests from all over the world (geographically distributed)
      - Response time required in order of few ms.

- DNS services
  1. hostname to IP addr translation
  2. host aliasing
  3. mail server aliasing
  4. Load distribution OR Load balancing

DNS mapping

www.xyz.com	$IP_1, IP_2, \dots IP_N$
⋮	⋮

DNS reply → $URL_1 = \frac{IP_1}{IP_4}$.

www.xyz.com

IP_1 S_1

IP_2 S_2

⋮ ⋮

IP_N S_N

List of IPs client picks one

DNS records

1. 'A' record : maps host names to IP address

{ www.iiitd.ac.in, xxx.xxx.xxx.xxx, A }

2. 'NS' record

name server

DNS req → $\boxed{\text{ac.in}}$ Auth. server

NS: { ac.in, iiitd.ac.in, NS }

A: { iiitd.ac.in, x.y.z.a, A }

DNS reply

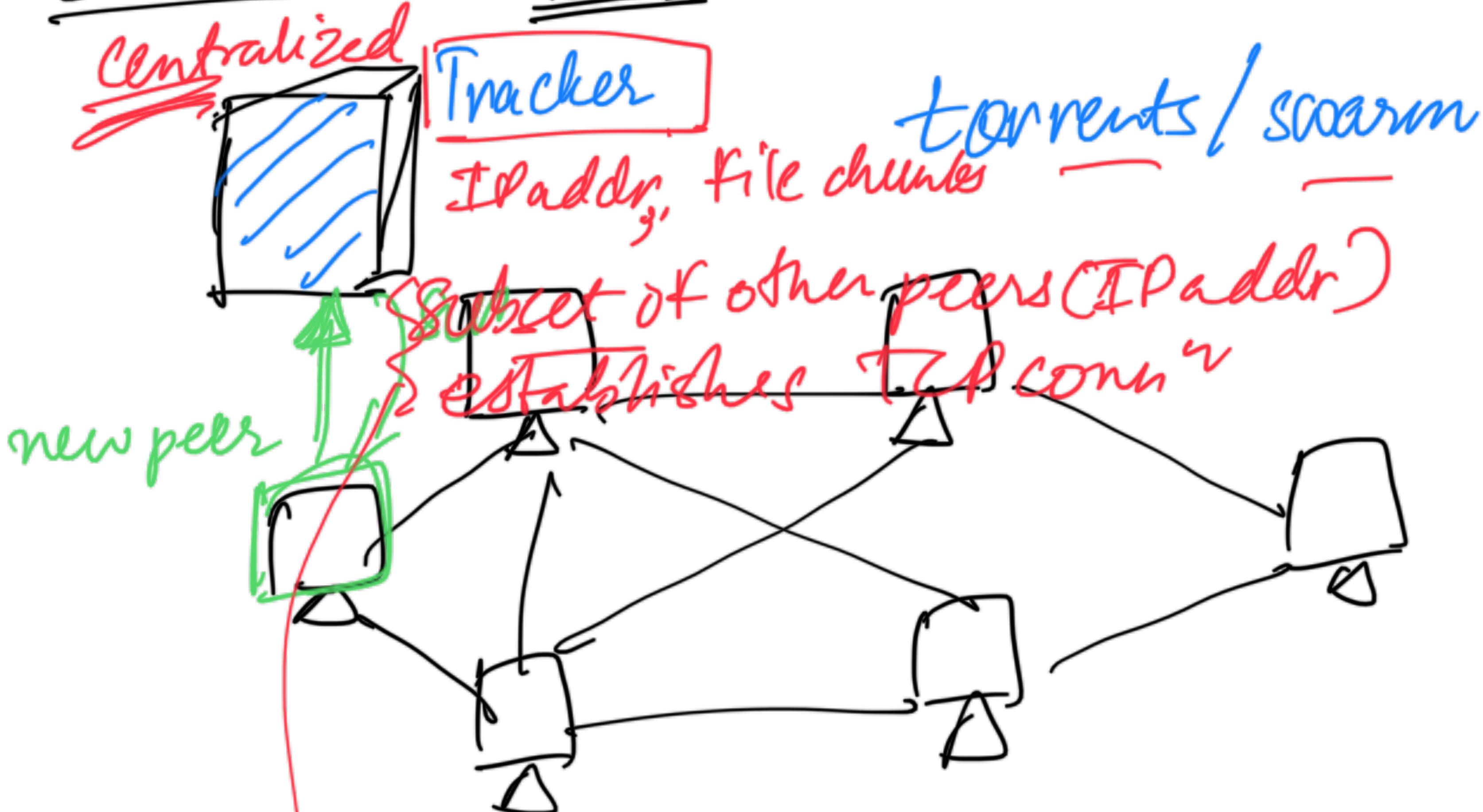
3. 'MX' record

4. 'CNAME' record

aliases

{ alias.name, actual.name, C

BitTorrent P2P



- File ⇒ divided into chunks
  say, eg, 256KB

$\frac{2560\,KB}{10\ chunks}$

- peer can download from other
- " " upload to other peer

1.
2.
3. peer provides a list of chunks
4. peer can request for a chunk
   Download
   ↳ rarest first policy

5. upload chunks to other peers

↳ TIT FOR TAT policy

4 highest download rate peers

"unchoked"

Challenge –

Tracker becomes a bottleneck

→ distributed tracker

$\boxed{\text{Distributed Hash Table (DHT)}}$